

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 2-Mercaptoethanol

**CAS No** 60-24-2

**Synonyms** 2-Hydroxy-1-ethanethiol; Thioglycol

**Product Code** **L04030**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

No hazards identified

#### Health hazards

|                                                      |                            |
|------------------------------------------------------|----------------------------|
| Acute Oral Toxicity                                  | Category 3                 |
| Acute Dermal Toxicity                                | Category 2                 |
| Acute Inhalation Toxicity - Vapors                   | Category 3                 |
| Skin Corrosion/Irritation                            | Category 2                 |
| Serious Eye Damage/Eye Irritation                    | Category 1                 |
| Skin Sensitization                                   | Category 1 Sub-category 1A |
| Reproductive Toxicity                                | Category 2                 |
| Specific target organ toxicity - (repeated exposure) | Category 2                 |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 2 |

#### Label Elements



Skull and Crossbones



Health Hazard



Corrosion



Environment

**Signal Word****Danger****Hazard Statements**

H310 - Fatal in contact with skin  
 H315 - Causes skin irritation  
 H317 - May cause an allergic skin reaction  
 H318 - Causes serious eye damage  
 H361 - Suspected of damaging fertility or the unborn child  
 H373 - May cause damage to organs through prolonged or repeated exposure  
 H400 - Very toxic to aquatic life  
 H411 - Toxic to aquatic life with long lasting effects  
 H301 + H331 - Toxic if swallowed or if inhaled  
 Combustible liquid

**Precautionary Statements**

P201 - Obtain special instructions before use  
 P202 - Do not handle until all safety precautions have been read and understood  
 P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
 P262 - Do not get in eyes, on skin, or on clothing  
 P264 - Wash face, hands and any exposed skin thoroughly after handling  
 P270 - Do not eat, drink or smoke when using this product  
 P271 - Use only outdoors or in a well-ventilated area  
 P272 - Contaminated work clothing should not be allowed out of the workplace  
 P280 - Wear protective gloves  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
 P310 - Immediately call a POISON CENTER or doctor  
 P330 - Rinse mouth  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
 P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
 P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse  
 P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
 P405 - Store locked up  
 P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Stench  
 Toxic to terrestrial vertebrates  
 Toxicity to Soil Dwelling Organisms

## Section 3 - Composition and Information on Ingredients

| Component                | CAS No    | Weight % |
|--------------------------|-----------|----------|
| 2-Mercaptoethanol        | 60-24-2   | <100     |
| Ethanol, 2,2'-dithiobis- | 1892-29-1 | <0.25    |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                                      |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                      |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                         |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                          |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                   |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                    |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Causes severe eye damage. May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                              |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Sulfur oxides, Sulfides.

### Decomposition Temperature

> 157°C

### Specific Hazards Arising from the Chemical

Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors. Do not allow run-off from fire-fighting to enter drains or water courses.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                    |
| <b>Respiratory Protection</b>   | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b> | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>  | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                     |                                          |
|------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Light yellow                        |                                          |
| <b>Physical State</b>                          | Liquid                              |                                          |
| <b>Odor</b>                                    | Stench                              |                                          |
| <b>Odor Threshold</b>                          | No data available                   |                                          |
| <b>pH</b>                                      | 5.2                                 | (0.1M)                                   |
| <b>Melting Point/Range</b>                     | -100 °C / -148 °F                   |                                          |
| <b>Softening Point</b>                         | No data available                   |                                          |
| <b>Boiling Point/Range</b>                     | 157 - 158 °C / 314.6 - 316.4 °F     |                                          |
| <b>Flash Point</b>                             | 73 °C / 163.4 °F                    | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available                   |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                      | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 2.3<br><b>Upper</b> 18 |                                          |
| <b>Vapor Pressure</b>                          | 1 mmHg @ 20 °C                      |                                          |
| <b>Vapor Density</b>                           | 2.69                                | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.110                               |                                          |
| <b>Bulk Density</b>                            | Not applicable                      | Liquid                                   |
| <b>Water Solubility</b>                        | Soluble                             |                                          |
| <b>Solubility in other solvents</b>            | No information available            |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                     |                                          |
| <b>Component</b>                               | <b>log Pow</b>                      |                                          |
| 2-Mercaptoethanol                              | -0.056                              |                                          |
| Ethanol, 2,2'-dithiobis-                       | -0.3                                |                                          |
| <b>Autoignition Temperature</b>                | 295 °C / 563 °F                     |                                          |
| <b>Decomposition Temperature</b>               | > 157°C                             |                                          |
| <b>Viscosity</b>                               | 3.43 cP at 20 °C                    |                                          |
| <b>Explosive Properties</b>                    |                                     | explosive air/vapour mixtures possible   |

**Oxidizing Properties** No information available

**Other information**

**Molecular Formula** C2 H6 O S  
**Molecular Weight** 78.13

## Section 10 - Stability and Reactivity

**Reactivity** None known, based on information available

**Stability** Stable under normal conditions. Water reactive.

**Conditions to Avoid** Incompatible products, Heat, flames and sparks, Excess heat, Exposure to moist air or water, Keep away from open flames, hot surfaces and sources of ignition.

**Incompatible Materials** Oxidizing agent, Acids, Strong bases.

**Hazardous Decomposition Products** Sulfur oxides. Sulfides.

**Hazardous Polymerization** No information available.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

**(a) acute toxicity;**

|                   |            |
|-------------------|------------|
| <b>Oral</b>       | Category 3 |
| <b>Dermal</b>     | Category 2 |
| <b>Inhalation</b> | Category 3 |

| Component                | LD50 Oral                | LD50 Dermal                                        | LC50 Inhalation |
|--------------------------|--------------------------|----------------------------------------------------|-----------------|
| 2-Mercaptoethanol        | LD50 = 244 mg/kg ( Rat ) | 150 µL/kg ( Rabbit )<br>112 - 224 mg/kg ( Rabbit ) |                 |
| Ethanol, 2,2'-dithiobis- | LD50 = 173 mg/kg ( Rat ) |                                                    |                 |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Sub-category 1A                                                  |

|                      |                          |
|----------------------|--------------------------|
| <b>Sensitization</b> | No information available |
|----------------------|--------------------------|

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Category 2

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Category 2

Target Organs Liver, Heart.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing:  
Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms.

| Component         | Freshwater Fish | Water Flea                                | Freshwater Algae                                  | Microtox                                  |
|-------------------|-----------------|-------------------------------------------|---------------------------------------------------|-------------------------------------------|
| 2-Mercaptoethanol |                 | EC50: = 1.52 mg/L, 48h<br>(Daphnia magna) | EC50: = 12 mg/L, 72h<br>(Desmodesmus subspicatus) | = 125 mg/L EC50<br>Pseudomonas putida 17h |

**Persistence and Degradability** Not readily biodegradable  
**Persistence** Persistence is unlikely.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.  
**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component                | log Pow | Bioconcentration factor (BCF) |
|--------------------------|---------|-------------------------------|
| 2-Mercaptoethanol        | -0.056  | No data available             |
| Ethanol, 2,2'-dithiobis- | -0.3    | No data available             |

**Mobility** The product is water soluble, and may spread in water systems. : Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN2966  
 Proper Shipping Name THIOGLYCOL  
 Hazard Class 6.1  
 Packing Group II

**ADG**

UN-No UN2966  
 Proper Shipping Name THIOGLYCOL  
 Hazard Class 6.1  
 Packing Group II

| Component                             | Hazchem Code |
|---------------------------------------|--------------|
| 2-Mercaptoethanol<br>60-24-2 ( <100 ) | 2X           |

**IATA**

UN-No UN2966  
 Proper Shipping Name THIOGLYCOL  
 Hazard Class 6.1  
 Packing Group II

**Environmental hazards** Dangerous for the environment  
 Product is a marine pollutant according to the criteria set by IMDG/IMO

**Special Precautions** No special precautions required

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                   | Standard for the Uniform Scheduling of Medicines and Poisons |
|-----------------------------|--------------------------------------------------------------|
| 2-Mercaptoethanol - 60-24-2 | Schedule 6 listed - in preparations for use as insect lures  |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                            | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------------------------|-------------------------------------------------------------|------------------------|
| 2-Mercaptoethanol - 60-24-2          | Present                                                     | -                      |
| Ethanol, 2,2'-dithiobis- - 1892-29-1 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable



**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|--------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| 2-Mercaptoethanol        | X    | X     | 200-464-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-23095 |
| Ethanol, 2,2'-dithiobis- | X    | X     | 217-576-6 | -      | X    | -   | X    | X     | -    |      | X     | KE-12729 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component                | CAS No    | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 2-Mercaptoethanol        | 60-24-2   | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Ethanol, 2,2'-dithiobis- | 1892-29-1 | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable

**Section 16 - Other Information****Legend**

**AICS** - Australian Inventory of Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

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Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Training Advice

Chemical incident response training.

#### Revision Date

18-Nov-2022

#### Revision Summary

SDS sections updated, 2, 3.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet